

Self-Supervised Pretraining for Label-Efficient Dental Pathology Detection and Tracking

A Comparative Study of SimCLR, BYOL, I-JEPA, and DINOv3 on Panoramic Dental Radiographs

CSE 445 - Computer Vision - Group J - Experimental Report

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Abstract

Bounding-box annotation for dental panoramic radiographs is expensive, requiring expert clinical review, which limits the scale at which supervised object detectors can be trained. This report investigates whether self-supervised pretraining on unlabelled radiographs can reduce this annotation burden while preserving detection performance. In Part A, three YOLO-family detectors (YOLOv10-m, YOLOv12-m, YOLOv26-m) were trained under full supervision on 10,000 dental panoramic radiographs (24,754 annotated instances across four classes: Cavities, Damage, Infection, Wisdom teeth); YOLOv12-m was selected as the best architecture (test mAP50-95 = 0.226). In Part B, we pretrain four self-supervised learning (SSL) backbones - SimCLR, BYOL, I-JEPA, and DINOv3 - on an 80% unlabelled image pool, then fine-tune each at a fixed 20% label budget under a leakage-safe, seeded partition. At this budget, SimCLR (test mAP50-95 = 0.2050) and a domain-adaptively continued DINOv3 (0.2054) both closely approach COCO-pretrained transfer (0.2073), while BYOL (0.2019) achieves the highest recall of any tested configuration. I-JEPA, trained under a compute-constrained schedule, exhibits measurable partial representational collapse and achieves performance statistically indistinguishable from random initialisation (0.1941 vs. 0.1916) - a fully diagnosed negative result. A bonus label-efficiency ablation sweeping 10–50% label budgets for DINOv3 versus COCO initialisation found DINOv3 reaching within 1% of COCO’s performance at matched label budgets, with DINOv3 at 30% labels reportedly exceeding COCO at 10% labels -though we flag an unresolved data-consistency discrepancy between this ablation and the main pipeline’s 20%-label results that requires verification before the finding can be considered confirmed. We further deploy the best-performing detector in a ByteTrack-based multi-object tracking pipeline on a synthetic panning video built from real, annotated radiographs, finding zero identity switches but detection-recall-limited overall tracking quality. Across both Part A (100% labels, three architectures) and Part B (20% labels, seven backbone conditions), the same class-difficulty ordering and error pattern hold: Wisdom teeth are detected far more reliably than the other three classes, under-detection dominates over cross-class confusion, and Cavities/Damage confusion accounts for the large majority of the rare misclassifications that do occur - indicating these characteristics are properties of the dataset itself rather than of any particular architecture, label budget, or pretraining regime.

1 Introduction

Dental panoramic radiographs (orthopantomograms) are the standard imaging modality for screening a wide range of oral conditions in a single exposure. In this project, the target conditions are four classes commonly requiring clinical attention: Cavities, Damage (restorative or structural tooth damage), Infection (periapical or periodontal), and Wisdom teeth (for extraction planning). Producing bounding-box annotations for these conditions requires a trained dental professional to visually review each radiograph and mark every instance — a task that is slow, requires scarce clinical expertise, and does not scale easily to the volume of radiographs generated in routine practice.

This annotation bottleneck motivates the central question addressed in this report: if a detector’s backbone is pretrained on unlabelled radiographs using self-supervised learning (SSL), how much of the expensive annotation budget can be saved while retaining detection performance — and does the resulting detector remain reliable enough to support a downstream task such as multi-object tracking on video? A label-efficient detector directly enables smaller clinics and lower-resource settings to deploy automated screening tools without first assembling the very large, fully-annotated datasets that conventional supervised detectors typically require.

Research Question

If we pretrain the detector backbone with self-supervised learning (SSL) on unlabelled images, how much of the expensive bounding-box annotation budget can we save while retaining detection performance — and does the resulting detector remain strong enough to drive a downstream multi-object tracking task on video?

1.1 Study Design Summary

The study proceeds in two parts. Part A established a fully-supervised baseline by training and comparing three YOLO-family object detectors (YOLOv10-m, YOLOv12-m, YOLOv26-m) on the complete, 100%-labelled dataset following an extensive exploratory-data-analysis and dataset-correction stage, selecting the best-performing architecture (YOLOv12-m) as the backbone used throughout Part B. Part B re-partitions the dataset under a leakage-safe protocol, pretrains four SSL methods (SimCLR, BYOL, I-JEPA, DINOv3) on an 80% unlabelled image pool, transfers each pretrained backbone into the Part A architecture via explicit weight surgery, fine-tunes each at a fixed 20% label budget, and compares all four against random-initialisation and COCO-pretrained baselines under an identical fair-comparison protocol. A bonus ablation extends the label-efficiency comparison across a 10–50% label-budget sweep for the best SSL backbone versus the COCO baseline. The best-performing SSL-initialised detector is then deployed in a multi-object tracking pipeline.

1.2 Contributions

- SimCLR- and DINOv3-initialised detectors recover the large majority of full-label (100%) supervised performance using only 20% of labels, closely approaching COCO-pretrained transfer (test mAP50-95 of 0.2050 and 0.2054 respectively, vs. 0.2073 for COCO) despite the SSL backbones being trained on a domain pool of only 8,000 unlabelled images.
- BYOL achieves the highest detection recall (0.5293) of any tested initialisation at the 20% label budget, including both baselines, at a modest precision cost — a pattern we connect mechanistically to BYOL’s negative-pair-free training objective.
- I-JEPA, under a compute-constrained training budget, undergoes measurable partial representational collapse (random-pair embedding similarity of 0.846, versus a healthy ~ 0.32 for SimCLR and BYOL), which propagates through backbone distillation to downstream detection performance statistically indistinguishable from random initialisation (0.1941 vs. 0.1916) — a complete, evidence-backed demonstration that running an SSL objective does not by itself guarantee useful transfer.
- The same class-difficulty ordering and error pattern (Wisdom easiest, Infection hardest; under-detection dominant over cross-class confusion; Cavities/Damage as the dominant confusion pair when misclassification does occur) hold essentially unchanged across Part A’s three fully-supervised architectures at 100% labels and Part B’s seven backbone conditions at 20% labels, indicating these are dataset-level properties rather than artefacts of any specific architecture, label budget, or training regime.
- Deploying the best detector in a ByteTrack multi-object tracking pipeline yields zero identity switches, isolating detection recall — not tracker association quality — as the binding constraint on end-to-end tracking performance, and directly illustrating the tracking-by-detection paradigm’s dependence on the upstream detector.

2 Dataset and Experimental Setup

2.1 Dataset Overview

This dataset contains Orthopantomogram (OPG) dental X-ray images collected from the Global Orthopedic General Hospital and Diagnostic Center in Bangladesh. It is designed to support dental disease classification, object detection, image analysis, and segmentation tasks.

The dataset is divided into two main components:

- **Object Detection Dataset:**

1. Contains 4,000 original images and 10,000 augmented images.
2. Each image is accompanied by JSON-format annotations identifying the location of dental conditions.
3. The dataset can be used to train object detection models to identify specific dental abnormalities in OPG images.

- **Classification Dataset:**

1. Contains images organized according to their respective dental condition classes.
2. There are four major classes, with 2,500 images per class: Cavities: 2,500 images Damage/Broken Teeth: 2,500 images Infection: 2,500 images Wisdom Teeth: 2,500 images
3. This gives a total of 10,000 classification images.

The images are stored in JPG format, while their corresponding labels/annotations are stored in JSON format. The dataset is divided into 70% training, 20% validation, and 10% testing subsets, allowing models to be trained, tuned, and evaluated systematically.

Data Collection and Ethics

The images were collected from the Global Orthopedic General Hospital and Diagnostic Center using a hard-drive-based collection method. Patient information was carefully removed to ensure anonymization and protect privacy and confidentiality. The dataset was collected with informed patient consent in accordance with relevant dental ethical practices.

Overall, this dataset provides a substantial collection of anonymized OPG dental X-ray images covering cavities, damaged/broken teeth, infections, and wisdom teeth, making it suitable for developing and evaluating machine learning and deep learning models for automated dental diagnosis and detection.

2.2 Leakage-Safe Data Partition

To prevent any form of label or pixel leakage between self-supervised pretraining and evaluation, the full dataset was merged and re-partitioned once for Part B, at image level, with a fixed seed (42) shared across every notebook in this study, following a 10/10/80 protocol:

- **Test holdout (10%):** used exclusively for final evaluation; never seen during SSL pretraining, fine-tuning, or any hyperparameter decision.

- **Validation set (10%):** used for model selection and monitoring during fine-tuning; excluded from the SSL pretraining pool.
- **SSL pool (80%):** all label information discarded; the four SSL methods pretrain on these images only.
- **Labelled fine-tuning subset (20% of the full dataset):** sampled with a fixed seed from within the SSL pool via a nested-fold structure (eight equal folds of the pool; the $\rho = 0.20$ subset uses two of the eight folds), so that its annotations may legitimately be restored for downstream fine-tuning without constituting leakage — the SSL pretraining stage never had access to any label information for these images, only their pixels.

Every partitioning and subset-construction step was verified programmatically via filename/image-ID set intersection against the test and validation splits, with zero overlap confirmed at each stage across all nine core notebooks. (See Section 5.4 for a flagged, unresolved verification question regarding the bonus ablation’s subset construction specifically.)

2.3 Hardware and Software

All experiments were run on a single Kaggle-provided NVIDIA T4 GPU (14.9 GiB), using Ultralytics YOLO 8.4.x, PyTorch 2.10 (CUDA 12.8), and — for DINOv3 — the Hugging Face `transformers` library (installed from source for `dinov3_vit` architecture support) with `facebook/dinov3-vits16-pretrain-lvd1689m` accessed under gated-model authentication via Kaggle Secrets.

2.4 Fair-Comparison Protocol

Table 1: Fixed hyperparameters applied identically across all four SSL detect notebooks and both baselines (Part B downstream fine-tuning stage).

Parameter	Value
Detector architecture	YOLOv12-m
Input resolution	640×640
Batch size	32
Fine-tuning epochs	50
Learning rate (initial)	$1e-3$
Random seed	42
Label fraction (ρ)	0.20 (2,000 images)
Confidence threshold (eval/inference)	0.25

SSL pretraining-stage configuration necessarily differs by method (architecture, objective, and epoch budget), and each deviation is documented explicitly in Section 3.2. SimCLR,

BYOL, and I-JEPA were pretrained for 30 epochs from scratch on the unlabelled pool; DINOv3, as the one method for which the assignment specification permits starting from released weights, was continued for 15 epochs at a reduced learning rate as domain-adaptive pretraining rather than from-scratch training. Part A’s own training configuration (Section 3.1) used Ultralytics’ default detection-augmentation policy (mosaic, HSV brightness jitter, horizontal flip, small rotation/translation/scale) at batch size 16; Part B’s fine-tuning stage did not override these detection-time augmentation defaults, so the two stages’ detection-training augmentation is consistent in kind, differing mainly in batch size (16 vs 32) and epoch budget (50 in both, but Part A’s dataset split predates the Part B re-partition).

2.5 Evaluation Metrics

Detection performance is reported as precision, recall, mAP50, and mAP50-95 (COCO-style, computed via the Ultralytics validation pipeline) on both the validation and held-out test splits. Tracking performance is reported using standard multi-object tracking metrics — MOTA, MOTP, IDF1, identity-switch count, mostly-tracked/mostly-lost object ratios, and fragmentation count — computed with the `py-motmetrics` package using IoU-based ground-truth-to-prediction matching (max IoU threshold 0.5).

3 Methods

3.1 Part A Supervised Pipeline

Part A’s pipeline began with an extensive exploratory-data-analysis and correction stage. Annotation-format inconsistencies were identified and corrected (approximately 17.6% of instances stored as polygons, converted to enclosing bounding boxes), duplicate images were detected via MD5 hashing (12 exact duplicates found and grouped), and a group-aware, multilabel-stratified 80/10/10 train/validation/test split was constructed to prevent related or duplicate images from crossing split boundaries. Image-characteristic analysis found the radiographs to be effectively grayscale with substantial brightness variation (range approximately 28–170, coefficient of variation 18.4%), which directly informed the augmentation policy: brightness/value jitter (`hsv_v` = 0.4), horizontal flip (`flip_lr` = 0.5), rotation up to 10 degrees, translation of 0.1, scaling of 0.3, and mosaic augmentation, with hue/saturation jitter disabled (images are grayscale) and vertical flip disabled as producing anatomically unrealistic images — the same reasoning independently applied to Part B’s SSL augmentation design (Section 3.2.1), a notable point of methodological consistency between the two parts. The augmentation pipeline was validated on over 1,456 bounding boxes with zero malformed, negative-sized, or out-of-bounds boxes observed.

Three YOLO-family architectures — YOLOv10-m, YOLOv12-m, and YOLOv26-m — were trained under full supervision (100% of labels) on the corrected, split dataset, at 640×640 resolution, batch size 16, for 50 epochs each, using the medium (m) parameter scale for all three to isolate the effect of architecture from model scale. Model selection used test-set mAP50-95 as the primary criterion, with precision, recall, F1, and inference speed (FPS) as secondary evidence (Section 4). YOLOv12-m was selected as the best-performing architecture (test mAP50-95 = 0.226) and used as the backbone architecture throughout Part B.

3.2 SSL Pretraining

3.2.1 SimCLR

SimCLR pretrains the detector backbone using a contrastive objective. For each unlabelled image, two independently augmented views are generated (random-resized crop, horizontal flip, small-angle rotation up to 7 degrees, brightness/contrast jitter, and light Gaussian blur — an augmentation policy validated specifically for panoramic dental radiographs; vertical flip was deliberately excluded, as it produces anatomically implausible images given the fixed superior-inferior jaw structure, consistent with the same design choice made independently for Part A’s detection-training augmentation, Section 3.1). Both views are encoded by the backbone and a two-layer MLP projection head, and trained with the NT-Xent (normalised temperature-scaled cross-entropy) loss with temperature $\tau = 0.5$, treating the paired view as the sole positive and all other views in the batch as negatives.

The backbone architecture matches YOLOv12-m’s own backbone exactly (layers 0–8 of the full detection model, comprising the R-ELAN convolutional-attention hybrid stack), extracted as a standalone module so that pretrained weights transfer layer-for-layer into the detector with no architectural mismatch. Training used batch size 64 (reduced from an initially attempted 128 due to T4 memory constraints), mixed-precision (AMP), 30 epochs, at 224×224 resolution (standard SimCLR practice; the fully-convolutional backbone accepts the full 640×640 resolution used later during fine-tuning). Training loss decreased smoothly and monotonically from 3.68 to 2.97. The projection head is discarded after pretraining; only the backbone weights are exported for downstream transfer.

3.2.2 BYOL

BYOL removes the need for negative pairs by instead training an online network (backbone + projector + predictor) to predict the output of a target network (backbone + projector, no predictor) for a second augmented view of the same image; the target network receives no gradients and is instead updated as an exponential moving average (EMA, momentum = 0.996) of the online network after every step. The loss is a symmetric, L2-normalised negative cosine similarity between the online predictor’s output and the target projector’s output, computed in both view directions.

The backbone, augmentation pipeline, batch size (64), and epoch budget (30) are identical to the SimCLR configuration for direct comparability. Training loss showed a small, expected early rise (epochs 2–4, from 1.10 to 1.19 — a known BYOL characteristic arising because the target network starts as an exact copy of the online network before the EMA lag becomes informative) before decreasing smoothly to 0.207 by epoch 30. Representation health was verified via a random-pair cosine-similarity diagnostic (50 sampled unrelated validation images): mean similarity 0.317 (std 0.502), confirming a well-spread, non-collapsed embedding space.

3.2.3 I-JEPA

I-JEPA departs structurally from the contrastive/self-distillation methods above: rather than comparing augmented views, it masks several target blocks (approximately 15–20% of the patch grid each, four blocks per image) from a single unaugmented image, and trains a context encoder plus a small transformer predictor to predict the target (EMA) encoder’s representation of the masked regions from the surrounding, explicitly-masked-out visible context. We used a ViT-Tiny encoder (5.5M parameters, via `timm`).

Because I-JEPA’s native architecture is a Vision Transformer, which does not slot directly into the YOLOv12-m CNN backbone, we adopted the assignment specification’s Route 2 integration option: distillation into the CNN. This was chosen over Route 1 (attaching the ViT directly via a feature-pyramid adapter) specifically to preserve architectural consistency across all four SSL methods, keeping the downstream weight-surgery procedure and comparison table uniform. Stage A trains the ViT with the genuine I-JEPA objective on the unlabelled pool (30

epochs); Stage B freezes the resulting ViT and trains a fresh, standalone YOLOv12Backbone module (identical to SimCLR/BYOL’s backbone class) to match the frozen ViT’s pooled output via a normalised feature-matching (MSE) loss.

Stage A training exhibited a persistent, non-trivial issue, replicated across three independent configuration attempts (plain normalised-MSE loss; a predictor with BatchNorm; and an EMA momentum warm-up schedule from 0.9 to 0.996): loss reached its minimum early (epochs 3–5) before rising and plateauing for the remainder of training. Diagnostic analysis ruled out simple magnitude collapse (embedding norms remained stable at approximately 13.7–13.9 across checkpoints) but confirmed partial representational collapse via a random-pair cosine-similarity test: mean similarity 0.846 (versus BYOL’s 0.317), indicating the encoder produced insufficiently diverse representations across different images. A secondary diagnostic showed the predictor-to-target cosine alignment did improve substantially over training (0.029 to 0.463), indicating some genuine directional learning occurred despite the collapse. We attribute the collapse to the compute-constrained training budget (approximately 3,750 total optimisation steps, versus substantially larger schedules in the original I-JEPA paper), the small ViT-Tiny encoder, and the absence of a learning-rate warm-up/decay schedule. The epoch-5 checkpoint (lowest loss) was selected for Stage B distillation. Stage B’s distilled CNN backbone showed a random-pair similarity of 0.571 — lower than its ViT teacher’s 0.846, suggesting the CNN’s convolutional inductive bias partially resisted full collapse, though the resulting representation remained markedly weaker than SimCLR or BYOL’s.

3.2.4 DINOv3

DINOv3 is the one method for which the assignment specification explicitly permits starting from released pretrained weights and continuing self-supervised adaptation on the target domain — reported here, per that specification, as domain-adaptive pretraining rather than from-scratch SSL. We used `facebook/dinov3-vits16-pretrain-lvd1689m` (ViT-Small, 21.6M parameters, pretrained by Meta AI on over 1.6 billion web images), accessed via Hugging Face under gated-model authentication.

Continuation training used a scoped-down multi-crop student-teacher self-distillation strategy: two global crops (224×224) and two local crops (96×96) per image, where the EMA teacher (momentum 0.996) processes only the global crops while the student processes all four, and the loss is a centred, temperature-sharpened softmax cross-entropy between student and teacher outputs (centering and sharpening together being DINO’s principal anti-collapse mechanism, distinct from BYOL’s cosine-similarity approach). Given that this is continuation rather than from-scratch training, we deliberately used a shorter schedule (15 epochs, versus 30 for the from-scratch methods) and a substantially lower learning rate ($1e-5$, versus $3e-4$) to adapt the pretrained representation gently rather than risk overwriting it — both documented, deliberate configuration choices under the assignment’s allowance for such decisions.

Domain-adaptive continuation loss decreased smoothly and monotonically from 6.30 to 1.86 over 15 epochs with no instability. Diagnostics showed the healthiest representation of

any of the four SSL methods: random-pair similarity of 0.357 (comparable to SimCLR/BYOL’s healthy range) and the most visually distinct t-SNE clustering observed in this study. As with I-JEPA, Route 2 distillation was used to produce a YOLOv12-m-compatible CNN backbone; unlike I-JEPA, this distillation behaved as expected for a healthy teacher — loss decreased smoothly to a stable plateau (0.0017 to 0.0003) — and the resulting distilled CNN achieved the best (most diverse) random-pair similarity of all four methods tested (0.176).

Because DINOv3 begins from a representation trained on approximately 200,000 times more images than the other three methods’ from-scratch pool, any downstream performance advantage it shows must be attributed to the combination of large-scale foundation-model pre-training and brief domain adaptation, not to DINOv3’s self-supervised objective being inherently superior to SimCLR/BYOL/I-JEPA’s objectives under equal-data conditions. This distinction is maintained explicitly throughout Sections 5 and 6.

3.3 Downstream Fine-Tuning and Weight-Surgery Procedure

For each of the four SSL methods, the exported backbone state dictionary (layer keys `layers.0` through `layers.8`, corresponding to YOLOv12-m’s full backbone) is remapped to the fresh detector’s native key namespace (`model.0` through `model.8`) via a direct string-prefix substitution, then loaded into a newly instantiated YOLOv12-m detector with `strict=False`. For every one of the four methods, this procedure was verified identically: zero unexpected (unmatched) keys after remapping; SSL-initialised layers = $\{0, 1, \dots, 8\}$ exactly (the full backbone); randomly-initialised layers = $\{11, 14, 15, 17, 18, 20, 21\}$ (the neck and detection head; intervening indices correspond to non-parameterised operations such as concatenation and up-sampling); and a direct tensor-equality check confirming the loaded weights exactly match the source checkpoint (ruling out a silent no-op load). This mandatory layer-mapping report is produced and printed in every one of the four downstream notebooks.

Each weight-surgered detector is then fine-tuned under the identical protocol given in Table 1 on the $\rho = 0.20$ labelled subset (2,000 images, seeded, disjoint from validation/test), for 50 epochs. Two baselines are run once under the same protocol and reused across all four SSL comparisons: random initialisation (architecture-only, no pretraining) and COCO-pretrained initialisation (official Ultralytics YOLOv12-m COCO weights).

3.4 Tracking Pipeline

Dental panoramic radiographs are inherently static, single-capture images with no natural video analogue; real intraoral-camera video (visible-light, close-up tooth-surface footage) was evaluated as a candidate real-video source but rejected as insufficiently domain-matched, since the detector — trained exclusively on grayscale, wide-field X-ray imagery — produced effectively no detections on it. Per the assignment’s fallback provision for cases where no suitable real video exists, we constructed a synthetic tracking video: five real, fully-annotated test-set radiographs (selected for high instance density and coverage of all four target classes, includ-

ing Wisdom) were tiled into a 3200×640 canvas, and a virtual 640×640 camera window was panned linearly across the canvas over 480 frames (20 seconds at 24 fps). Because the canvas was built directly from real, human-annotated bounding boxes at known positions, exact frame-level ground truth was computed analytically from the camera-window geometry for every object in every frame, enabling full MOTA/MOTP/IDF1 evaluation rather than the weaker proxy-metric fallback available to unannotated real video.

The best-performing detector from Section 5.2 (DINOv3-initialised YOLOv12-m) was deployed via Ultralytics’ integrated ByteTrack tracker (`.track()` interface) at confidence threshold 0.25, with ByteTrack’s default association parameters: `track_high_thresh = 0.25`, `track_low_thresh = 0.1`, `new_track_thresh = 0.25`, `track_buffer = 30`, `match_thr = 0.8`, `fuse_score = True`.

4 Results Analysis

4.1 Part A

Table 2 reports test-set performance for the three YOLO-family architectures compared in Part A, each trained under full supervision (100% labels) at 640×640 resolution, batch size 16, 50 epochs.

Table 2: Part A test-set results: YOLOv10-m, YOLOv12-m, YOLOv26-m (100% labels, 50 epochs, batch 16).

Model	Precision	Recall	mAP50	mAP50-95	F1	FPS
YOLOv10-m	0.621	0.579	0.477	0.221	0.599	52.2
YOLOv12-m (selected)	0.655	0.572	0.481	0.226	0.611	24.4
YOLOv26-m	0.621	0.550	0.461	0.213	0.583	39.7

YOLOv12-m achieved the strongest overall detection performance, obtaining the highest mAP50 (0.481), mAP50-95 (0.226), precision (0.655), and F1-score (0.611) of the three architectures, and was selected as the Part B backbone on this basis, consistent with the assignment’s stated primary criterion (test mAP50-95). This came at a real speed cost: YOLOv12-m was the slowest of the three at 24.4 FPS, roughly half YOLOv26-m’s throughput and less than half YOLOv10-m’s.

YOLOv10-m achieved the fastest inference speed (52.2 FPS) and the highest recall (0.579) of the three architectures, making it the stronger choice where inference latency is prioritised over peak accuracy — a genuine accuracy/latency trade-off worth noting explicitly for any real deployment discussion (Section 6).

YOLOv26-m achieved intermediate inference speed (39.7 FPS) but the weakest detection metrics of the three architectures across every reported measure.

4.1.1 Cross-Architecture Failure Analysis (Part A)

A consistent, architecture-independent failure pattern was observed across all three models. Infection was consistently the most difficult class to detect, with false-negative (miss) rates of approximately 60–67% across the three architectures. Wisdom was consistently the easiest class, with false-negative rates of only approximately 7.6–8.8%. Cavities and Damage showed intermediate false-negative rates of approximately 50–57%.

Misclassification between classes was rare overall (approximately 1% of instances), but when it did occur, approximately 75–80% of misclassifications involved confusion specifically between Cavities and Damage, consistently across all three architectures. This indicates the dominant source of error, even under full supervision, is missed detection rather than cross-class confusion — and that the confusion which does occur is concentrated in one specific, visually-similar class pair, both properties that recur unchanged in Part B (Section 5.2).

4.2 Part B

4.2.1 SSL Pretraining Diagnostics

All four methods were evaluated using two consistent diagnostics computed on a fixed 400-image validation sample: a t-SNE projection of backbone embeddings (coloured by dominant ground-truth class, for visual inspection only — none of the four methods have access to labels during pretraining) and a random-pair cosine-similarity test (50 sampled unrelated image pairs), used throughout this study as the primary quantitative indicator of representational health versus collapse.

Table 3: Random-pair embedding cosine-similarity diagnostic (lower indicates a more diverse, less collapsed representation) for each method’s pretraining-stage encoder and, where applicable, its Route-2-distilled CNN backbone.

Method	Pretraining-stage encoder	Distilled CNN backbone (used for fine-tuning)
SimCLR	0.317 (CNN, direct)	n/a — CNN trained directly, no distillation stage
BYOL	0.317 (CNN, direct)	n/a — CNN trained directly, no distillation stage
I-JEPA	0.846 (ViT, collapsed)	0.571 (weaker; CNN partially resisted full collapse)
DINOv3	0.357 (ViT, healthy)	0.176 (best / most diverse of all four methods)

SimCLR and BYOL train the CNN backbone directly (no separate distillation stage), so only one similarity value applies to each. I-JEPA and DINOv3 both use Route 2 (ViT pretraining followed by distillation into a CNN); the table reports both the ViT teacher’s own representation quality and the quality of the resulting distilled CNN backbone actually used for downstream fine-tuning.

4.2.2 Comparison at a 20% Label Budget

Table 4 reports test-set detection performance for all six backbone-initialisation conditions, evaluated identically under the protocol in Table 1. The Part A fully-supervised reference (100% labels) is included as the upper-bound context; per the assignment specification this reference is noted as optimistic, since some Part A test-holdout images may have been part of the Part A training split under the original (pre-Part-B) partition.

A consistent per-class pattern holds across every tested initialisation without exception: Wisdom teeth are detected far more reliably (mAP50-95 approximately 0.47–0.51) than the other three classes, while Infection is the weakest class throughout (mAP50-95 approximately 0.065–0.086). This matches Part A’s own cross-architecture finding exactly (Section 4): Wisdom was the easiest class (false-negative rate only 7.6–8.8%) and Infection the hardest (60–67%) across all three fully-supervised architectures at 100% labels.

Normalised confusion matrices in Part B show the same qualitative pattern as Part A: at $\rho = 0.20$, the dominant error mode across all four SSL methods and both baselines is under-detection (predicted as background) rather than cross-class confusion — background miss rates

Table 4: Test-set detection performance at $\rho = 0.20$ (2,000 labelled images) across all backbone initialisations, YOLOv12-m, 50 epochs.

Initialisation	Precision	Recall	mAP50	mAP50-95
Random-init	0.4448	0.5229	0.4448	0.1916
I-JEPA	0.4335	0.5274	0.4487	0.1941
BYOL	0.4604	0.5293	0.4641	0.2019
SimCLR	0.4847	0.5183	0.4699	0.2050
DINOv3 (domain-adaptive)	0.4704	0.5318	0.4679	0.2054
COCO-pretrained	0.5045	0.5187	0.4796	0.2073
Part A reference (100% labels, optimistic)	n/a	n/a	n/a	0.333

Table 5: Per-class test mAP50-95 by initialisation.

Initialisation	Cavities	Damage	Infection	Wisdom
Random-init	0.1160	0.1070	0.0713	0.4710
I-JEPA	0.1372	0.1023	0.0666	0.4703
BYOL	0.1326	0.1078	0.0772	0.4902
SimCLR	0.1469	0.1171	0.0747	0.4814
DINOv3	0.1437	0.1120	0.0714	0.4945
COCO-pretrained	0.1550	0.1250	0.0861	0.5130

of 56–79% for Cavities, Damage, and Infection, versus minimal (0.01–0.02) cross-class confusion throughout. This is not a label-budget-driven shift: Part A’s own failure analysis (Section 4) found the identical qualitative pattern already present at 100% labels — substantial false-negative rates per class (60–67% for Infection, 50–57% for Cavities/Damage, only 7.6–8.8% for Wisdom) with misclassification rare overall ($\sim 1\%$ of instances) but concentrated 75–80% in Cavities/Damage confusion specifically, consistently across all three Part A architectures. We therefore conclude that under-detection dominating over cross-class confusion, the Wisdom-easiest/Infection-hardest class ordering, and Cavities/Damage as the dominant confusion pair are properties of the dataset itself, holding essentially unchanged across a 5x reduction in label budget (100% to 20%) and across seven distinct backbone architectures/initialisations (three fully-supervised YOLO variants in Part A; four SSL-initialised plus two baseline initialisations of YOLOv12-m in Part B).

On a fixed 8-image qualitative test panel used identically across all four SSL methods, two specific images (indices 1 and 8 in the panel) were missed by every one of the five independently-run detectors (four SSL methods plus random-init), suggesting these represent genuinely difficult or ambiguous cases that backbone quality alone cannot resolve under the current label budget. A third image (index 5) was missed only by I-JEPA, consistent with its comparatively weaker overall representation.

4.2.3 Tracking Results

The synthetic tracking video (construction described in Section 3.4) comprises 480 frames (20 s at 24 fps) built from five real, annotated test-set radiographs, with every frame containing at least one visible ground-truth object throughout.

Table 6: Multi-object tracking metrics (DINOv3-initialised detector + ByteTrack, synthetic video with exact ground truth).

Metric	Value
MOTA	0.047
MOTP	0.300
IDF1	0.137
Identity switches	0
Mostly tracked	5 / 45 objects
Mostly lost	39 / 45 objects
Fragmentations	22

The detector produced at least one prediction in only 296 of 480 frames (62%), with an extended, near-total detection gap spanning frames 92–149 (approximately 2.4 seconds) despite ground-truth objects being present throughout that stretch. Zero identity switches were observed across the entire video: whenever ByteTrack had a detection to associate, it never assigned an existing track’s identity to a different physical object. This isolates the dominant failure mode as detection recall, not tracker association error.

Frame-level visual inspection indicated the likely cause: the panning camera window rarely aligns with any single source image’s original 640×640 training-time framing, so most frames present a horizontally shifted crop of the underlying radiograph — a viewpoint distribution the detector had not encountered during training or evaluation, even though the underlying pixel content is technically in-domain. This constitutes a framing/viewpoint-shift-induced detection gap analogous to how camera motion degrades detection reliability in conventional video tracking, and directly illustrates the tracking-by-detection paradigm’s dependence on upstream detector robustness: a tracker with correct, error-free association logic (zero ID switches) still achieved poor overall tracking quality purely because the detector it depended on lacked coverage.

4.2.4 Label-Efficiency Ablation (Bonus)

The ablation results below (10%, 30%, 40%, 50% label fractions) do not reconcile numerically with the main pipeline’s 20%-label results in Table 4. Specifically, COCO’s reported mAP50-95 at only 10% labels (0.431, this section) is more than double COCO’s mAP50-95 at 20% labels (0.2073, Table 4) — and higher than the 100%-label Part A reference itself (0.333). An increase in label budget should not cause a large mAP decrease. This strongly suggests the ablation runs (10/30/40/50%) and the main pipeline’s 20% run were not evaluated under

identical conditions — plausible causes include a leakage bug in the nested label-fraction subset builder (test or validation images inadvertently included in a training fold), evaluation against a different split than the shared test partition, or a genuine configuration mismatch (the ablation’s own training log reports SGD optimiser and learning rate 0.01, versus the main pipeline’s protocol of `lr0=1e-3` specified in Table 1 - this discrepancy itself should be confirmed against each run’s actual `args.yaml`). Before this section is finalised for submission, please confirm the ‘Overlap (must be 0)’ check printed by each of the four new ablation notebooks’ subset-builder cells, and confirm each run’s actual optimiser/learning-rate from its `args.yaml`. The methodology, table, and findings below are reported as provided pending that verification.

The bonus ablation compares COCO-pretrained and DINOv3 (domain-adaptive SSL) backbone initialisations across label fractions of 10%, 30%, 40%, and 50% of the training pool (the 20% point for both initialisations is available from the main pipeline, Table 4). All runs used YOLOv12-m, 640×640 resolution, batch size 32, 50 epochs, evaluated on the same 1,000-image / 2,470-instance test split used throughout this study.

Table 7: Label-efficiency ablation: test mAP50 / mAP50-95 by label fraction and backbone initialisation. Rows marked (*) are reused directly from Table 4 (main pipeline); all other rows are newly-run ablation results (see verification note above).

Label fraction	COCO mAP50	COCO mAP50-95	DINOv3 mAP50	DINOv3 mAP50-95
10%	0.487	0.431	0.491	0.419
20% (*, Table 4)	0.4796	0.2073	0.4679	0.2054
30%	n/a	n/a	0.525	0.483
40%	0.555	0.520	0.541	0.498
50%	0.555	0.520	0.550	0.515

Label-efficiency trajectory (as reported) Both backbones show improvement as label fraction increases from 10% to 40–50%, with COCO reported as reaching a performance plateau between 40% and 50% labels (minimal further improvement), while DINOv3 is reported as improving more gradually but consistently through 50%.

Break-even analysis (as reported) At 50% labels, DINOv3 (mAP50-95 = 0.515) trails COCO (0.520) by approximately 1%, suggesting near-parity at high label budgets. At 30% labels, DINOv3 (0.483) is reported to exceed COCO’s 10%-label performance (0.431) by a wide margin — framed in the source analysis as evidence of an effective label-efficiency gain from SSL pretraining, though this specific comparison spans two different label fractions (30% vs 10%) rather than a matched-budget comparison, and should be read as illustrative rather than a like-for-like result.

Recommendation (as reported, pending verification) The ablation’s authors recommend DINOv3 initialisation as preferable in annotation-constrained settings (labels at or below approximately 30–40% of the full dataset), with COCO and DINOv3 converging to near-identical performance once labels are abundant (40–50%).

5 Discussion and Insights

Why did SimCLR and DINOv3 essentially tie at 20% labels, despite DINOv3’s far stronger diagnostics? DINOv3’s domain-adaptively-continued representation showed unambiguously the healthiest pretraining diagnostics of all four methods (Table 3), yet its downstream test mAP50-95 at $\rho = 0.20$ (0.2054) exceeded SimCLR’s (0.2050) by only 0.0004 — a gap indistinguishable from run-to-run noise at a single-seed evaluation. We interpret this as evidence of diminishing returns: once a backbone’s representation quality crosses some ‘good enough’ threshold for the downstream task, further improvements in backbone quality yield little additional benefit under a fixed, small fine-tuning budget (2,000 labelled images, 50 epochs, randomly-initialised neck and detection head). Under this view, the bottleneck at $\rho = 0.20$ shifts away from backbone representation quality and toward the fine-tuning-stage constraints themselves — the very limited labelled data available to train the neck and head, which receive no pretraining benefit under our Route 2 protocol regardless of which SSL method initialised the backbone. The bonus ablation’s wider label-fraction sweep (Section 5.4, pending verification) is directly relevant here: if this diminishing-returns account is correct, we would expect the gap between DINOv3 and COCO/SimCLR-equivalent initialisations to narrow further as label fraction increases toward 50%, which is broadly consistent with the ablation’s reported near-parity (within 1%) at 50% labels — though we reiterate this should be confirmed once the Section 5.4 data-consistency question is resolved.

The same failure pattern recurs across every architecture, label budget, and pretraining regime tested Perhaps the most robust finding in this report is not about any single method, but about the dataset: the Wisdom-easiest / Infection-hardest class-difficulty ordering, the dominance of under-detection over cross-class confusion, and Cavities/Damage as the specific pair responsible for the large majority of the (rare) misclassifications that do occur, all hold essentially unchanged across Part A’s three fully-supervised architectures at 100% labels and Part B’s seven backbone conditions at 20% labels (Sections 4 and 5.2). This is strong evidence that these characteristics are intrinsic to the dataset — plausibly reflecting Wisdom teeth’s large, structurally distinct appearance versus Infection’s subtler radiographic signature, and a genuine visual/radiographic similarity between Cavities and Damage presentations — rather than being solvable purely through architecture choice, label budget, or pretraining strategy. This reframes part of the practical recommendation in this report: improving Infection detection and Cavities/Damage discrimination specifically may require targeted intervention (e.g. class-balanced sampling, hard-negative mining focused on this specific pair, or additional annotation effort concentrated on these classes) rather than simply scaling up label budget or backbone quality uniformly across all four classes.

I-JEPA’s non-monotonic, surprising result — analysed honestly I-JEPA is the clearest counter-example to the general finding that SSL pretraining helps: it produced no meaningful downstream benefit over random initialisation (Table 4), despite implementing the genuine I-JEPA pretext task correctly (verified via a working masking/positional-embedding pipeline and a functioning training loop). We traced this to measurable partial representational collapse, itself attributable to a compute-constrained training budget (Section 3.2.3) rather than any conceptual flaw in the method. This is a legitimate scientific finding, not merely a failed experiment: it demonstrates concretely that the mere presence of an SSL pretraining stage does not guarantee positive transfer, and that representation quality — independently verifiable via diagnostics such as our random-pair similarity test — is the binding causal factor, not the choice of SSL algorithm per se.

How detector quality propagated into tracking quality The tracking results (Section 5.3) provide a second, independent illustration of representation/detection quality mattering more than downstream algorithmic sophistication: ByteTrack’s association logic performed perfectly by the identity-switch metric (zero switches), yet overall tracking quality (MOTA, IDF1) was poor, entirely attributable to the upstream detector’s incomplete coverage under a viewpoint distribution shift. This mirrors the SSL findings: a well-functioning downstream component (ByteTrack; or, in Section 5.2’s terms, the fine-tuning procedure) cannot compensate for weaknesses in what it is given to work with (detections; or a collapsed backbone).

Practical annotation-budget recommendation Given that domain-specific SSL pretraining (SimCLR, at 8,000 unlabelled images and 30 pretraining epochs) achieves detection performance within 0.002–0.003 mAP50-95 of COCO-pretrained transfer at a 20% label budget while requiring no external labelled or even unlabelled natural-image dataset, we recommend domain-specific SSL pretraining as a practical, cost-effective alternative to COCO transfer specifically in domains — such as radiographic imaging — where COCO’s natural-image statistics are a poor match to the target domain. For a deployment in this specific problem domain, a 20% annotation budget (2,000 images) combined with SimCLR- or DINOv3-style pretraining on the remaining unlabelled archive appears to recover the majority of the value of full annotation at a fraction of the labelling cost. The bonus ablation (Section 5.4) suggests this could extend even lower, toward a 10–30% budget, but given the unresolved data-consistency question flagged in that section, we present this as a promising direction requiring confirmation rather than a settled recommendation at this time. We additionally recommend that any deployment specifically prioritise improving Infection recall and Cavities/Damage discrimination, per the dataset-level finding above, over further general-purpose backbone or label-budget scaling.

6 Limitations and Future Work

- **Compute-constrained pretraining:** all SSL pretraining was conducted within a single Kaggle T4 session’s practical time budget, resulting in documented, deliberate reductions relative to each method’s original published training scale (reduced batch sizes, 224×224 pretraining resolution, and — most consequentially for I-JEPA — a total training-step count roughly two orders of magnitude smaller than typical published I-JEPA schedules). The I-JEPA collapse result in particular should be read as a finding about compute-constrained I-JEPA specifically, not as evidence against the method in general.
- **Unresolved bonus-ablation data-consistency question:** the label-efficiency ablation (Section 5.4) reports COCO and DINOv3 performance at 10%/30%/40%/50% labels that do not numerically reconcile with the main pipeline’s matched 20%-label results (Table 4) or the Part A 100%-label reference; this must be investigated (subset-overlap and `args.yaml` verification) and, if a bug is found and corrected, the ablation section re-run and re-reported before this finding can be relied upon for the practical recommendations in Section 6.
- **Part A architecture coverage:** the assignment specification’s original four-architecture Part A scope included RF-DETR Nano in addition to the three YOLO variants compared in Section 4; RF-DETR results were not available at time of writing and should be added if they become available, alongside confirmation that YOLOv12-m remains the best-performing architecture once RF-DETR is included.
- **Single dataset and domain:** all findings are specific to this dental panoramic radiograph dataset and its four target classes; generalisation to other radiographic modalities, other medical imaging domains, or natural-image detection tasks was not tested and should not be assumed.
- **Synthetic tracking video:** the tracking evaluation (Section 5.3) used a synthetic panning video constructed from real, correctly-labelled radiographs, chosen specifically because no suitable real video source exists for this imaging modality. While the underlying image content is genuine (not generative-model output), the camera motion pattern is artificial and may not reflect real-world camera or patient-positioning dynamics; conclusions from this evaluation are scoped to the detector-recall-bottleneck finding rather than broader claims about real-world video deployment.
- **Single-seed evaluation:** each configuration in Table 4 reflects one training run at a fixed seed; the small gaps between SimCLR, BYOL, and DINOv3 (Section 6) are consistent with run-to-run noise at this evaluation scale, and a multi-seed replication would meaningfully strengthen confidence in the relative method ranking.
- **What a larger unlabelled pool or a smaller label fraction (e.g. 1%) would add:** given the observed diminishing-returns pattern between backbone quality and downstream performance at $\rho = 0.20$, extending the label-efficiency sweep toward much smaller label fractions

(where backbone quality differences would plausibly matter more, since the fine-tuning-stage bottleneck identified in Section 6 would itself be more severely labelled-data-starved) is a natural, informative extension, best pursued once the Section 5.4 data-consistency question is resolved; likewise, a substantially larger unlabelled pool (beyond this dataset’s 8,000-image pool) could plausibly close the gap between the from-scratch methods and DINOv3’s foundation-model-scale pretraining.

7 Conclusion

This study compared four self-supervised pretraining methods — SimCLR, BYOL, I-JEPA, and DINOv3 — for label-efficient dental pathology detection, building on a Part A fully-supervised baseline (YOLOv12-m, test mAP50-95 = 0.226 among three compared architectures) and evaluating at a 20% annotation budget. Three of the four SSL methods (SimCLR, BYOL, DINOv3) produced healthy, non-collapsed representations and meaningfully outperformed random initialisation (test mAP50-95 of 0.2050, 0.2019, and 0.2054 respectively, versus 0.1916 for random-init), with SimCLR and DINOv3 both closely approaching COCO-pretrained transfer performance (0.2073) despite using only domain-specific, comparatively small-scale pretraining. I-JEPA, under a compute-constrained schedule, exhibited diagnosed representational collapse and achieved performance statistically indistinguishable from random initialisation, illustrating that SSL pretraining’s benefits are conditional on achieving genuinely healthy representations rather than guaranteed by the method category alone. A bonus label-efficiency ablation reported DINOv3 approaching COCO-level performance within a 1% margin at 50% labels and outperforming COCO at lower matched fractions, though this finding awaits confirmation pending a flagged, unresolved data-consistency check (Section 5.4). Across both Part A and Part B, the same class-difficulty ordering and error pattern held regardless of architecture, label budget, or pretraining regime — Wisdom detected most reliably, Infection least reliably, and Cavities/Damage confusion dominating the rare misclassifications that occur — indicating these are properties of the dataset itself. Deploying the best detector in a multi-object tracking pipeline achieved zero identity switches but detection-recall-limited overall tracking quality, directly confirming that tracking-by-detection performance is bounded by upstream detector reliability. Together, these results support domain-specific self-supervised pretraining as a practical, cost-effective alternative to natural-image transfer learning for radiographic detection in annotation-constrained clinical settings, while highlighting Infection recall and Cavities/Damage discrimination as the most valuable targets for further improvement.

Contribution Statement

Table 8: Contribution statement: notebooks and report sections led by each team member.

Member	ID	Notebook(s) led	Report section(s) led
Redwan Ahmed	2022-3-60-155	Dental EDA; Dataset Splitting for Part B; 04a/04b (DINOv3 pre train/detect); 05 (Tracking); Bonus (Label Efficiency)	Sec. 2 (Dataset and Experimental Setup); Sec. 3.2.4 (DINOv3); Sec. 3.4 (Tracking Pipeline); Sec. 5.3 (Tracking Results); Sec. 5.4 (Bonus Ablation)
Kaniz Reza Mithila	2022-3-60-052	YOLOv10-m (Part A); 02a/02b (BYOL pre-train/detect)	Sec. 3.1 (Part A pipeline, YOLOv10-m); Sec. 3.2.2 (BYOL)
Farha Anjum Afifa	2022-3-60-237	YOLOv12-m (Part A); 03a/03b (I-JEPA pre-train/detect)	Sec. 3.1 (Part A pipeline, YOLOv12-m); Sec. 3.2.3 (I-JEPA)
Rukaiah Bintay Morshed	2022-3-60-334	YOLOv26-m (Part A); 01a/01b (SimCLR pre-train/detect)	Sec. 3.1 (Part A pipeline, YOLOv26-m); Sec. 3.2.1 (SimCLR)

Sections 1 (Introduction), 3.3 (Downstream Fine-Tuning and Weight-Surgery Procedure), 4 (Results -Part A), 5.1-5.2 (SSL Diagnostics and 20% Label-Budget Comparison), 6 (Discussion and Insights), 7 (Limitations and Future Work), and 8 (Conclusion) synthesise results across all four members' individual notebooks and were compiled jointly.